Research Report

2026 S.T. Yau High School Science Award (Asia)

Bundle Neural Networks for Functional Connectomes

A Gauge-Aware, Capacity-Controlled Test of Density-Dependent Aggregation

Team Member	Adit Patil
School	School of Scholars
City and Country	Akola, India
Registration Number	Comp-183
Supervising Teacher	Anjali Tupe
Position	Class Teacher
Supervising School	School of Scholars, Akola, India
Report Version	ABIDE-I operator study with revision analyses, 15 August 2026

This report contains the frozen ABIDE-I operator comparison and three separately labelled post-hoc studies. It does not report external validation or additional architecture experiments.

The source code, analysis records, and accepted evidence snapshots are kept in the accompanying project repository.

Abstract

Functional connectomes are often treated as graphs, but a common node feature already contains one region's correlations with all 116 regions in the atlas. Message passing may therefore average information that is already global. I tested whether the orthogonal transport in Bundle Neural Networks (BuNNs) changes this density-dependent aggregation behavior. The study used 754 technically eligible ABIDE-I participants from 18 sites. Five neural variants shared one two-layer backbone: identity propagation, a learned-local capacity control, GCN aggregation, trivial-bundle heat diffusion, and learned BuNN transport. Graphs retained the strongest positive edges at 1, 5, 10, and 20 percent density. Model selection used nested held-out-site evaluation, and the analysis included regularized non-graph baselines and common-frame representation diagnostics.

The primary BuNN-minus-GCN density-curve contrast was -0.00958 (95% site-bootstrap confidence interval [-0.03665, 0.01146]). The participant-weighted estimate changed sign to +0.0030, but its interval also included zero. BuNN was below the connectome elastic net by -0.05516 (95% interval [-0.08297, -0.02752]). Its matched-anchor effective-rank contrast was -0.00719 (95% interval [-0.01310, -0.00105]), which did not support the proposed representation-preservation advantage.

Post-hoc analyses examined the null result. An RBF-SVM reached 0.6255 equal-site balanced accuracy but did not exceed the 0.6401 elastic-net result. In an audit of 360 accepted BuNN checkpoints, replacing learned maps with random orthogonal maps reduced balanced accuracy by 8.86 percentage points (95% interval $[-12.36, -4.78]$; Holm-adjusted exact $p = 0.0038$). A separate 700-cell synthetic study gave the models known topology and recoverable local frames. BuNN gained 42.75 points on the specified conditional contrast (95% interval $[38.42, 47.67]$), but the advantage disappeared when topology or frame information was broken.

The learned maps changed BuNN's predictions, and BuNN solved a controlled task that required transport. Neither result produced a detected advantage in the specified ABIDE-I pipeline. The study evaluates a computational operator, not biological bundle geometry or a clinical diagnostic system.

Keywords: functional connectome; bundle neural network; graph neural network; representation collapse; held-out-site validation; ABIDE-I; autism

Acknowledgments and research assistance

This study used public, anonymized ABIDE-I derivatives distributed by the Preprocessed Connectomes Project. The ABIDE investigators and participating institutions collected and shared the original data but were not involved in this analysis. Neural training ran on an AWS EC2 instance with an NVIDIA A10G GPU. The software stack included Python, PyTorch, NumPy, pandas, scikit-learn, and scientific plotting libraries.

OpenAI Codex was used frequently from 31 July to 15 August 2026 for literature leads, protocol criticism, coding and debugging, AWS run monitoring, analysis checks, test construction, chart and table preparation, structural editing, language polishing, and document formatting. Its work affected the repository, analysis scripts, figures, tables, and this revision draft. The detailed dated record is in `docs/ai_use_log.md`. Product records must be checked for the exact Codex model/version before submission.

Anthropic Claude was used for methodological criticism of the proposal, and Cursor was used for interactive coding assistance, as reported by the student. Their versions, dates, frequency, retained chats, and exact affected tasks have not been supplied and are not inferred in this report.

Author-only items required before submission. The student must add the missing Claude and Cursor details, state whether prior supervisor approval covered each AI system, describe the supervising teacher's actual contribution, attach the required chat records, and independently verify every formula, citation, result, and paragraph. The official rules prohibit AI ghostwriting of the report body. This assisted revision cannot be submitted as independently written student work unless the student rewrites and owns the academic expression in accordance with those rules.

Commitments on academic honesty and integrity

We hereby declare that we:

1. are fully committed to the principle of honesty, integrity and fair play throughout the competition;
2. perform the research work ourselves and thus truly understand the content of the work;
3. observe the common standard of academic integrity adopted by most journals and degree theses;
4. have declared all assistance and contributions received from any person, agency, or institution for the research work;
5. have truthfully disclosed all AI tool usage in the research report and obtained prior approval from the supervising teacher, and have submitted all supporting documentation required by the competition;
6. avoid contact with assessment-panel members that may create a direct or indirect conflict of interest;
7. avoid interaction with assessment-panel members that could undermine their neutrality or the fairness of the assessment;
8. observe the safety regulations of every venue used for experiments, where applicable;
9. observe all competition rules and regulations; and
10. agree that the decision of YHSA(Asia) is final in all matters related to the competition.

We understand that failure to honor these commitments may lead to disqualification or removal of an award, and that unethical conduct may be reported to the school principal and other relevant parties.

Student signature: ______________________ Date: __________

Supervising teacher signature: ______________________ Date: __________

Noted and endorsed by school principal:

Principal signature: ______________________ Date: __________

Principal name: ______________________ School: School of Scholars

Contents

1 Introduction

Resting-state functional magnetic resonance imaging records changes in the blood-oxygen-level-dependent signal while a participant performs no assigned task. A functional connectome summarizes correlations between regional time series. Brain regions are represented as nodes and their correlations as weighted edges, which makes graph neural networks appear well suited to the data.

That choice assumes message passing will add useful information. In many connectome classifiers, however, a node already receives a full row of the correlation matrix. Its feature vector describes its association with every region in the atlas. Averaging neighboring rows may remove distinctions the classifier needs without introducing any missing context.

Controlled benchmarks have found that classical and non-graph models can match or outperform graph deep-learning models on functional-connectome tasks. Their ablations identify aggregation as one possible source of the loss [1]. Earlier multi-site ABIDE work also showed that atlas, preprocessing, and validation choices can materially change unseen-site performance [2]. A graph representation alone is therefore not enough to justify propagation. Any proposed operator should be compared on the same splits with no propagation and with strong regularized baselines.

BuNNs use a different propagation rule. They learn orthogonal maps that align node features before diffusion [3]. Prior theory and synthetic experiments suggest this can limit representation collapse. Neural sheaf diffusion provides related evidence that non-trivial transport can change the behavior of message passing [4]. Whether that advantage survives in dense functional connectomes with globally informed node features remained unresolved.

1.1 Refining the research question

The original proposal treated negative functional correlations as a possible analogue of heterophilic message passing. That interpretation did not survive formal scrutiny. Correlation sign does not identify graph-learning heterophily, anatomical connectivity, neural excitation or inhibition, or causal flow. It also depends on preprocessing choices such as global-signal regression [5]. The study therefore retained ABIDE-I, the connectome representation, the BuNN architecture, and the aggregation problem, but moved the hypothesis to the level of the propagation operator.

The resulting study asks:

> On a pre-specified ABIDE-I pipeline, does learned bundle transport produce a more

favorable held-out-site performance curve across graph densities than GCN aggregation, and does any predictive pattern co-occur with stronger common-frame representation preservation?

The decision rule required BuNN to exceed GCN, matched no-propagation controls, and the strongest regularized connectome baseline. Merely degrading less than another graph operator did not count as an advantage.

2 Related work and rationale

2.1 ABIDE-I and cross-site prediction

ABIDE-I combined resting-state fMRI and phenotypic data from sites in several countries [6]. The Preprocessed Connectomes Project later released derivatives from pipelines that include C-PAC [7]. Scanners, acquisition protocols, participant composition, motion, and preprocessing vary between sites. Random participant splits can therefore put similar acquisition conditions in both training and test sets. Holding out a complete site asks whether a model generalizes to a data-collection setting it never saw during fitting.

2.2 Graph convolution and aggregation

A GCN combines a node's features with normalized neighboring features, then applies a learned transformation [8]. Repeated propagation or a denser graph can contract the resulting representations, although the effect depends on the data and architecture. Here each node begins with a complete connectivity row, so the no-propagation model is a full scientific control rather than a minor ablation.

2.3 Bundle transport and coordinate frames

BuNNs associate node features with local coordinate frames and learn orthogonal maps that align them before diffusion [3]. This complicates representation analysis. Embeddings from two nodes can look different simply because they use different frames. Stacking raw embeddings before calculating cosine similarity or effective rank could reward rotations rather than retained information. The analysis therefore moves embeddings to a common learned frame before comparing nodes. A learned-local control keeps the map generator but removes diffusion between nodes. Effective rank is useful here because rank collapse is one proposed component of over-smoothing, but it does not by itself establish that useful information was lost [9].

2.4 Why published ABIDE percentages are not one leaderboard

Published ABIDE graph results are often quoted together even though they use different prediction problems. Parisot et al. connected participants, not brain regions, and built edges from phenotypic information that included sex and acquisition site. Their 10-fold transductive evaluation produced 70.4% accuracy and 0.75 AUC on 871 participants [10]. Wang et al. reported 71.6% mean leave-one-site-out accuracy on 1,057 participants. Their cGCN used CCS-preprocessed CC200 time series, a training-derived k-nearest-neighbour graph, five graph-convolution layers, and temporal averaging [11]. Yang et al. reported 64.27% average accuracy for a graphlet-augmented GCN on 1,035 participants, as well as a 75.9% site-specific maximum [12].

These values cannot be ranked against the equal-site balanced accuracy in this report. The studies differ in cohort filtering, atlas, preprocessing, input representation, graph definition, split, metric, and model selection. Wang's released leave-one-site-out script also supplies the left-out site to Keras as validation data and uses that performance for stopping and checkpoint selection. The published result still describes the released procedure, but 71.6% is not a clean unseen-site target for the present experiment. Han et al. offer the closest conceptual comparison because their controlled benchmark also found that simple models often matched or exceeded graph models and that aggregation performance declined as density increased [1].

Table 1: Protocol context for selected ABIDE graph-model results. Reported percentages are retained in their original metrics and should not be ranked as if they came from one experiment.

Study	Data and representation	Evaluation and result	Main mismatch with this study
Present study	754; C-PAC no-GSR; AAL-116 connectivity-row brain graphs	Nested leave-one-site-out; equal-site BA; elastic net 0.6401, GCN curve 0.5945, BuNN curve 0.5849	Primary outcome is a balanced-accuracy density curve with test sites excluded from model selection.
Wang et al. [11]	1,057; CCS; CC200 raw ROI time series; training-derived k-NN graph	Leave-one-site-out ordinary accuracy; 71.6% at $k = 5$	Different input, atlas, architecture, cohort, and metric; released code uses the left-out site for model selection.
Parisot et al. [10]	871; participant population graph with imaging and phenotypic edges	10-fold transductive CV; 70.4% accuracy, 0.75 AUC	Nodes are participants; graph uses site and sex; folds mix sites.
Yang et al. [12]	1,035; functional graphs and graphlet counts	64.27% average accuracy; 75.9% highest site value	Different features and aggregation; a highest-site value is not an overall estimate.
Han et al. [1]	1,009; C-PAC; CC200 connectomes	AUROC benchmark and density ablations	Best used as a relative aggregation comparison, not an absolute score target.

3 Methods

3.1 Study organization

The design changed only the propagation operator while keeping the remaining pipeline fixed. Classical baselines were completed and audited before the five neural variants were run with identical inputs, outer test sites, grouped inner folds, candidate budgets, stopping rules, and final seeds. The complete neural output was checked for expected coverage before any scores were opened, and the confirmatory analysis was frozen at that point. The accepted run contained 9,324 fits, 52,780 held-out predictions, and all 18 outer sites.

Study 1 is the confirmatory ABIDE-I operator comparison. E1 and E2 were added after its result was known. E1 intervenes on the 360 accepted BuNN checkpoints; E2 tests the proposed computation on synthetic graphs whose topology and coordinate frames are known. They help interpret Study 1 but cannot serve as independent confirmation of a mechanism in ABIDE-I. Deeper networks, Wang-style time-series training, and ABIDE-II evaluation are outside the completed evidence.

3.2 Study 1: dataset and technical quality control

The source was the ABIDE-I phenotype table and the C-PAC band-pass-filtered AAL ROI time-series derivative without global-signal regression. The AAL atlas divides the brain into 116 macroscopic regions in standardized space [13]. The cohort was built in a fixed sequence. Of 1,112 phenotype rows, 77 lacked an available derivative file identifier, 38 were removed at the first manual QC gate, 221 at the manual functional QC gate, and 7 because their sites did not meet the analysis-site rule. This left 769 downloaded ROI files. Fifteen contained a zero-variance ROI, for which Pearson correlation is undefined. The final cohort contained 754 participants: 371 labelled ASD and 383 controls from 18 sites.

Mean framewise displacement was 0.124 among retained participants, 0.135 among records removed at the functional QC gate, and 0.212 among records removed at the earlier manual gate. These are descriptive summaries of available values. They do not show that motion caused a rating or that QC removed every form of site or sampling bias.

Technical eligibility does not make the cohort representative of a wider population. As figure 1 shows, sites differed in size and class balance. The evaluation retains that variation and accounts for it at the site level.

Table 2: Participant accounting from the public phenotype table to the frozen analysis cohort. The two manual gates and the zero-variance check are distinct steps.

Stage	n	ASD	Control	Removed
All phenotype rows	1112	539	573	0
Valid diagnosis	1112	539	573	0
Available FILE_ID	1035	505	530	77
Manual anatomical QC retained	997	476	521	38
Manual functional QC retained	776	379	397	221
Eligible 18-site set	769	378	391	7
Final nonzero-variance cohort	754	371	383	15

Table 3: Frozen cohort and feature summary.

Quantity	**Value**
Downloaded ABIDE-I ROI time-series files	769
Zero-variance-ROI technical exclusions	15
Technically eligible participants	754
ASD / control participants	371 / 383
Evaluable acquisition sites	18
AAL regions per participant	116
Unique undirected connectome features	6,670

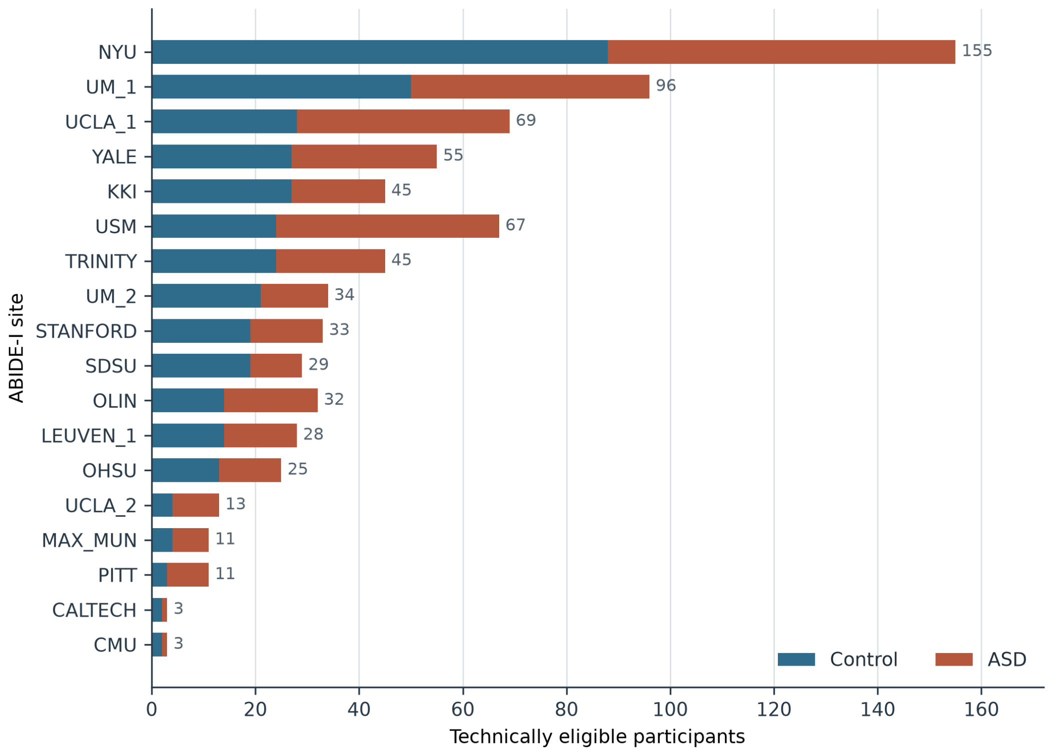

Figure 1: Participant composition of the frozen technically eligible cohort. Sites varied in both total size and class balance, motivating site-level evaluation and equal-site aggregation. This is a descriptive count figure, not a performance result.

3.3 Connectome construction

For each participant, every pair of ROI time series was correlated and then transformed with Fisher's r-to-z function. The diagonal was set to zero. The finite, symmetric matrix supplied 6,670 lower-triangle features to the classical models. In the neural models, each node received its complete 116-value connectivity row.

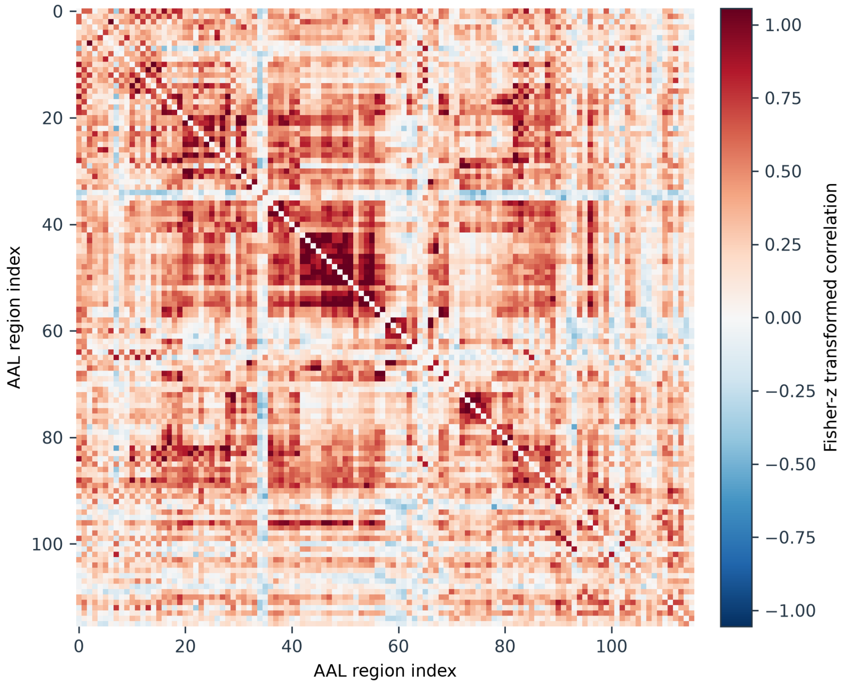

Figure 2: A deterministic example from the frozen archive illustrating a participant-level Fisher-z functional-connectivity matrix. Red and blue indicate positive and negative statistical associations. They are not interpreted as excitation, inhibition, anatomy, or causal flow.

At 1, 5, 10, and 20 percent density, each graph retained its strongest positive off-diagonal associations symmetrically. Density zero supplied the matched identity or learned-local anchor. Negative associations remained in the node profiles but were not propagation edges. This pre-specified construction was computational and carried no biological interpretation.

3.4 Classical baselines

Three logistic models served as practical baselines. A covariates-only model used age, sex, mean framewise displacement, and usable scan length. An elastic-net model used all vectorized connectome features, with combined ℓ_1 and ℓ_2 penalties for correlated high-dimensional predictors [14]. The third model combined those connectome features and covariates. Scaling, feature processing, and tuning used training sites only.

After Study 1, E1, and E2 were complete, a post-hoc nonlinear comparator was added.

An RBF-kernel support vector machine used the same 6,670 connectome features and frozen nested site splits. Each inner fit standardized its own training partition. The grid crossed $C \in \{0.1, 1, 10\}$ with $\gamma \in \{0.25/D, 1/D, 4/D\}$, where $D = 6670$. Candidates were selected by the unweighted mean balanced accuracy across inner validation sites. The decision threshold remained fixed at zero, and no probability calibration, PCA, feature selection, or harmonization was used. This comparison was not allowed to alter the confirmatory Study 1 decision.

3.5 Shared neural backbone and operator controls

All neural variants shared the encoder, hidden width, two propagation layers, nonlinearities, normalization, dropout, pooling, classifier, optimizer, training schedule, and tuning budget. Only the propagation rule, or the map capacity needed to isolate that rule, changed between variants.

Table 4: Neural operator and capacity controls.

Variant	Operation	Purpose
Identity	No inter-node propagation	Tests whether message passing adds value.
Learned-local	Learns the BuNN node-wise maps but does not diffuse between nodes	Separates map-generator capacity from transport.
GCN	Normalized neighborhood aggregation	Standard graph-diffusion reference.
Trivial bundle	Bundle diffusion with identity transport maps	Separates the diffusion construction from learned transport.
Learned BuNN	Learned orthogonal node maps followed by bundle diffusion	Tests the proposed transport operator.

The bundle hidden state had eight two-dimensional bundles and two channels per bundle, giving width 32. In each of the two layers, a feature-conditioned map network $32 \to 32 \to 8$ produced one angle for every node and bundle. Four bundles used rotations and four used reflections, so each map was exactly orthogonal. The two map networks added 2,640 parameters. Identity, GCN, and trivial-bundle models had 5,889 parameters; learned-local and learned BuNN had 8,529.

The bundle variants used the exact heat kernel

$$H = \exp(-tL), \qquad L = I - D^{-1}A, \qquad t = 1,$$

with zero Laplacian rows for isolated nodes. Learned BuNN moved each local field to its learned common frame, applied the shared linear update and heat diffusion, then returned it to the local frame. Learned-local performed the same map and update operations without heat diffusion between nodes.

3.6 Nested held-out-site evaluation

Each site served once as the outer test site. Within the remaining sites, four grouped folds selected among four equal-budget AdamW candidates using two tuning seeds. The selected configuration was then refitted with five fixed final seeds. Test-site data were excluded from scaling, tuning, stopping, and model selection. The primary endpoint was the unweighted mean of held-out-site balanced accuracy, so every acquisition site had equal influence. Pooled balanced accuracy and AUROC were secondary endpoints.

Table 5: Complete Study 1 model-selection, architecture, and training settings.

Model or component	**Tuned values**	**Fixed settings**
Covariates logistic	C: 0.01, 0.1, 1, 10	L2; balanced classes
Connectome elastic net	C: 0.03, 0.3, 3; l1 ratio: 0.1, 0.5, 0.9	SAGA; balanced classes
Combined elastic net	C: 0.03, 0.3, 3; l1 ratio: 0.1, 0.5, 0.9	SAGA; balanced classes
All neural operators	(learning rate, weight decay): (0.0003, 1e-05); (0.0003, 0.0001); (0.001, 1e-05); (0.001, 0.0001)	AdamW; batch 8; at most 150 epochs; patience 20
Shared neural backbone	2 propagation layers; width 32; dropout 0.2	GELU; node-mean pooling; one graph-level logit; 2 tuning and 5 final seeds
Bundle representation	Not tuned	8 bundles; dimension 2; 2 vector-field channels per bundle; $8 \times 2 \times 2 = 32$ hidden coordinates
Heat operator	Not tuned	$t = 1$; exact $\exp(-tL)$; $L = I - D^{-1}A$; isolated-node Laplacian rows fixed at zero
Map generator, each layer	Not tuned	MLP $32 \to 32 \to 8$ with GELU; one angle per node and bundle; 4 rotations and 4 reflections
Parameter totals	Not tuned	Identity, GCN, trivial bundle: 5,889; learned-local and BuNN: 8,529; map-generator difference: 2,640

3.7 Density-curve and representation estimands

The primary predictive estimand was the normalized trapezoidal area under the balanced-accuracy curve over the complete density grid. GCN and trivial-bundle curves used identity at zero density; the learned-BuNN curve used the parameter-matched learned-local anchor. Using the complete curve prevented a favorable density from being selected after the results were known. The five normalized trapezoid weights at 0, 1, 5, 10, and 20 percent were 0.025, 0.125, 0.225, 0.375, and 0.250. Thus 62.5 percent of the curve estimate came from the 10 and 20 percent points. A post-hoc uniform-grid average checked whether that weighting choice changed the result.

After each propagation layer, BuNN embeddings were moved into a common learned frame before nodes were compared. Normalized effective rank at layer 2 was the primary representation endpoint. Normalized dispersion, mean pairwise cosine similarity, invariant edge-transport distance, and the rank change from encoder to layer 2 were secondary. These measures describe representation behavior; they cannot establish that a representation change caused a prediction change.

3.8 Statistical analysis and decision rule

The five final seeds were averaged within each site and configuration before site-level inference. Confidence intervals came from 10,000 paired bootstrap resamples of the 18 held-out sites. Curve contrasts also used exact two-sided paired sign-flip tests. Density-specific tests were secondary and received Holm correction within each named family.

A positive architectural result required all three of the following:

1. stronger common-frame representation preservation than GCN;
2. better held-out prediction than GCN and the matched no-propagation controls; and
3. performance above the strongest regularized non-graph baseline.

When an interval contained zero, the result was reported as no detected advantage. The protocol also recorded a 0.03 balanced-accuracy reference margin for the equal-site curve. Because that margin was not supported by an external minimum-effect justification, it is used only to describe the scale of the interval, not to claim formal equivalence.

3.9 Revision-only sensitivity analyses

After the fitted-model analyses were complete, a separate revision protocol was frozen for calculations that required no retraining. It reconstructed the QC cascade, added equal-site and participant-weighted site-cluster bootstrap intervals, repeated the density comparison with uniform grid weights, and computed all six pairwise E1 intervention contrasts with Holm correction.

The same protocol characterized the implemented heat operator on every participant graph. The spectrum of $I - D^{-1}A$ was computed through the symmetric-similar matrix $I - D^{-1/2}AD^{-1/2}$, with the code's zero convention for isolated nodes. For each eigenvalue λ, the retained heat response was $e^{-\lambda}$ at $t = 1$. This analysis describes contraction in the actual graphs. It does not establish that contraction caused a classification error.

3.10 Run integrity and reproducibility

Long jobs wrote atomic checkpoints for each site and recorded hashes of the data, configuration, and source code. Site ownership records, restart validation, canonical merging, and independent completion audits protected the parallel run. Before unblinding, the neural artifact was checked for coverage, recomputed metrics, diagnostics, runtime records, and warnings. Reported values are generated from the accepted hash-bound snapshot rather than copied by hand.

3.11 E1: accepted-checkpoint intervention audit

After Study 1 was frozen, E1 tested which components the 360 accepted learned-BuNN checkpoints depended on, without refitting any model. Each checkpoint first had to reproduce its archived held-out probabilities within 10^{-6}. Four interventions were compared with unaltered inference: identity maps, learned maps shuffled between nodes, random orthogonal maps, and graph rewiring by double-edge swaps that preserved every participant's degree sequence. A permutation of encoded nodes was used only to check equivariance.

The randomized families used 100 stored randomizations shared across the five model seeds. Balanced accuracy was calculated for each seed and randomization. Randomizations were averaged within each site-density cell, followed by the seeds. The four density effects were then averaged within each site, and sites received equal weight. Every effect is reported as intervention minus unaltered BuNN. Uncertainty used 10,000 paired site-bootstrap samples. Exact two-sided sign-flip tests enumerated all 2^{18} site sign assignments, with Holm correction across the four primary interventions.

Secondary endpoints included absolute probability changes, classification flips, AUROC, sensitivity, specificity, and four final-layer gauge-aware diagnostics. Spearman correlations across the 72 site-density cells described the association between representation and prediction changes; they were not mediation tests. The complete run was archived before the analysis protocol and code were frozen. The analyzer passed synthetic known-answer tests before one recorded unblinding invocation, and an independent recomputation audit passed before interpretation.

3.12 E2: synthetic known-geometry study

E2 was frozen after Study 1 and E1, so it provides explanatory rather than confirmatory evidence about ABIDE-I. Each replicate contained 600 balanced synthetic graphs with 116 nodes and fixed train, validation, and test partitions of 360, 120, and 120 graphs. Class labels did not alter the marginal type of node values. Class 1 signals were smooth on a degree-six ring graph; class 0 used a participant-specific permutation of the same kind of smooth signal.

Seven operators shared a fixed identity encoder, one 32-dimensional propagation layer, GELU activation, mean pooling, optimizer, stopping rule, and test protocol. They were identity, GCN, trivial-bundle diffusion, fixed-random transport, learned-local, learned BuNN, and an oracle that received the true maps. The fixed encoder kept the planted coordinate frames identifiable. Learned-local retained BuNN's map generator without cross-node diffusion.

The condition families were no geometry (S0), fixed recoverable node frames (S1), incorrect topology (S2), shuffled frame markers (S3), graded marker corruption (S4), participant-specific frames without an informative marker (S5), and a global-feature analogue (S6). S4 used 15, 30, 60, and 120 degree corruption. Ten paired replicate seeds produced 700 cells. Test predictions remained hidden until every cell passed structural and hash audits and 100 condition-replicate groups were independently sealed.

The primary estimand was

$$[(\text{BuNN} - \text{GCN})_{S1} - (\text{BuNN} - \text{GCN})_{S0}],$$

measured in test balanced-accuracy percentage points. Support required three ordered gates: a positive oracle versus GCN contrast in S1; a primary mean of at least three points with a positive paired-bootstrap lower bound and exact sign-flip $p < 0.05$; and a positive BuNN versus learned-local lower bound in S1. A BuNN versus GCN result above three points in S0 triggered a capacity warning. The complete analysis was invoked once and independently recomputed.

The first full launcher was stopped before unblinding because a regression review found that half of its nominal S0 maps were reflections rather than identity maps. Version 2 used explicit identity matrices and added an exact regression test. Every cell was restarted without a version 1 checkpoint or prediction.

4 Results

4.1 Study 1: classical baseline performance

The connectome elastic net was the strongest classical model, with equal-site balanced accuracy of 0.6401. Adding covariates gave 0.6336, and covariates alone gave 0.5652. The connectome features contained cross-site predictive information in this pipeline. These scores measure computational prediction, not clinical diagnostic performance.

Table 6: Held-out-site performance of the classical baselines.

Model	Equal-site BA	Pooled BA	Pooled AUROC
Covariates-only logistic regression	0.5652	0.5629	0.5973
Connectome elastic net	**0.6401**	**0.6495**	**0.6794**
Connectome + covariates elastic net	0.6336	0.6428	0.6773

4.2 Predictive performance across graph density

Performance declined with density for every diffusion operator (figure 3). The normalized curve-area difference between learned BuNN and GCN was -0.00958 (95% confidence interval [-0.03665, 0.01146]; exact $p = 0.524$). Because the interval crossed zero, the analysis detected no predictive advantage. Its upper bound was 0.0115, below the protocol's 0.03 reference margin. This describes the precision of the equal-site curve contrast; it is not a formal equivalence claim. Smaller effects and other BuNN designs remain possible.

Relative to the site-matched connectome elastic net, the learned-BuNN curve difference was -0.05516 (95% confidence interval [-0.08297, -0.02752]; exact $p = 0.0018$). The entire interval was below zero. Learned BuNN also failed the positive-result rule against identity and learned-local controls.

Table 7: Pre-specified predictive contrasts, reported as balanced-accuracy differences.

Contrast	Estimate	95% CI	Exact p
BuNN curve − GCN curve	−0.0096	[−0.0366, 0.0115]	0.524
BuNN curve − trivial-bundle curve	−0.0062	[−0.0293, 0.0108]	0.681
BuNN nonzero mean − learned-local	−0.0167	[−0.0347, 0.0009]	0.096
BuNN nonzero mean − identity	−0.0146	[−0.0320, 0.0018]	0.116
BuNN curve − connectome elastic net	−0.0552	[−0.0830, −0.0275]	0.0018

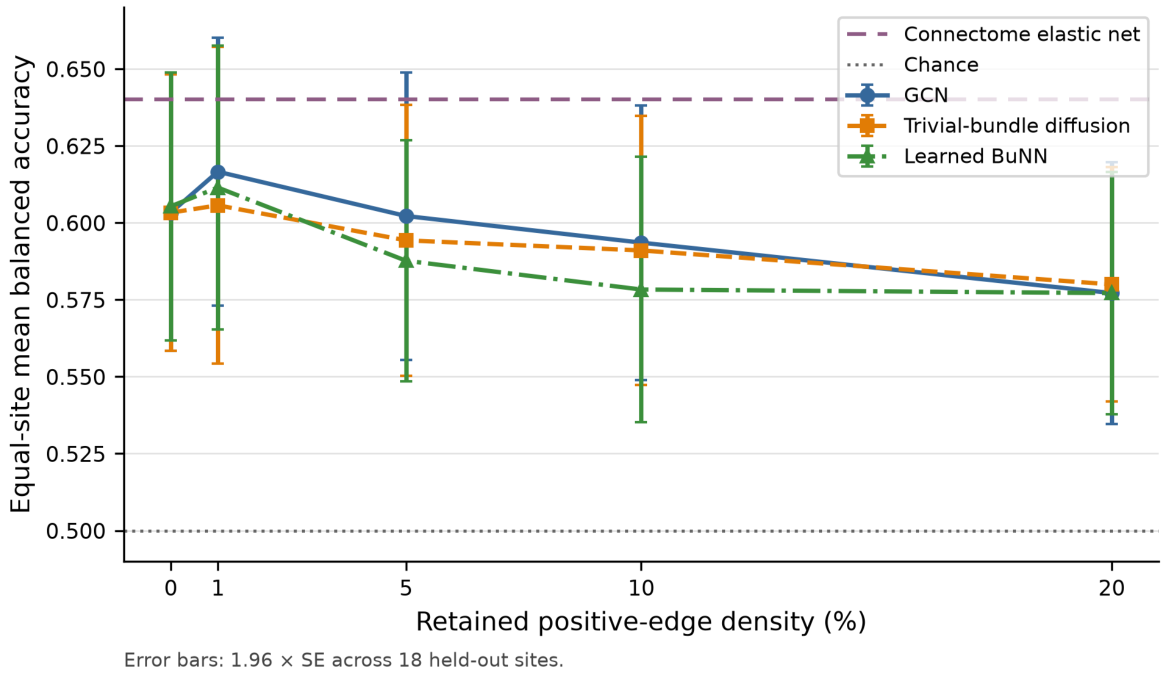

Figure 3: Equal-site mean held-out balanced accuracy across the pre-specified positive-edge density grid. Error bars show 1.96 standard errors across sites for visualization. Confirmatory inference used paired site bootstrap and exact sign-flip tests on the locked curve estimand.

4.3 Common-frame representation behavior

Before matching anchors, BuNN appeared to have a higher effective-rank curve than GCN. Its learned-local zero-density value, however, was already high. After each operator's matched anchor was subtracted, the normalized effective-rank contrast was -0.00719 (95% confidence interval [-0.01310, -0.00105]). The direction opposed the proposed BuNN preservation advantage. Normalized dispersion and pairwise cosine similarity followed the same pattern.

The learned-local control changes the interpretation. Without it, diversity created by the additional map generator could be mistaken for information preserved through inter-node transport. Once the matched anchor is used, the results do not support that explanation. The diagnostics are computational measurements and cannot identify a biological mechanism.

Table 8: BuNN-minus-GCN matched-anchor representation contrasts.

Layer-2 endpoint	Estimate	95% CI
Normalized effective rank (primary)	−0.0072	[−0.0131, −0.0010]
Normalized dispersion	−0.0417	[−0.0523, −0.0306]
Mean pairwise cosine similarity	0.0415	[0.0319, 0.0507]
Invariant edge-transport distance	−0.6436	[−0.8205, −0.4720]
Encoder-to-layer-2 effective-rank change	−0.0056	[−0.0144, 0.0048]

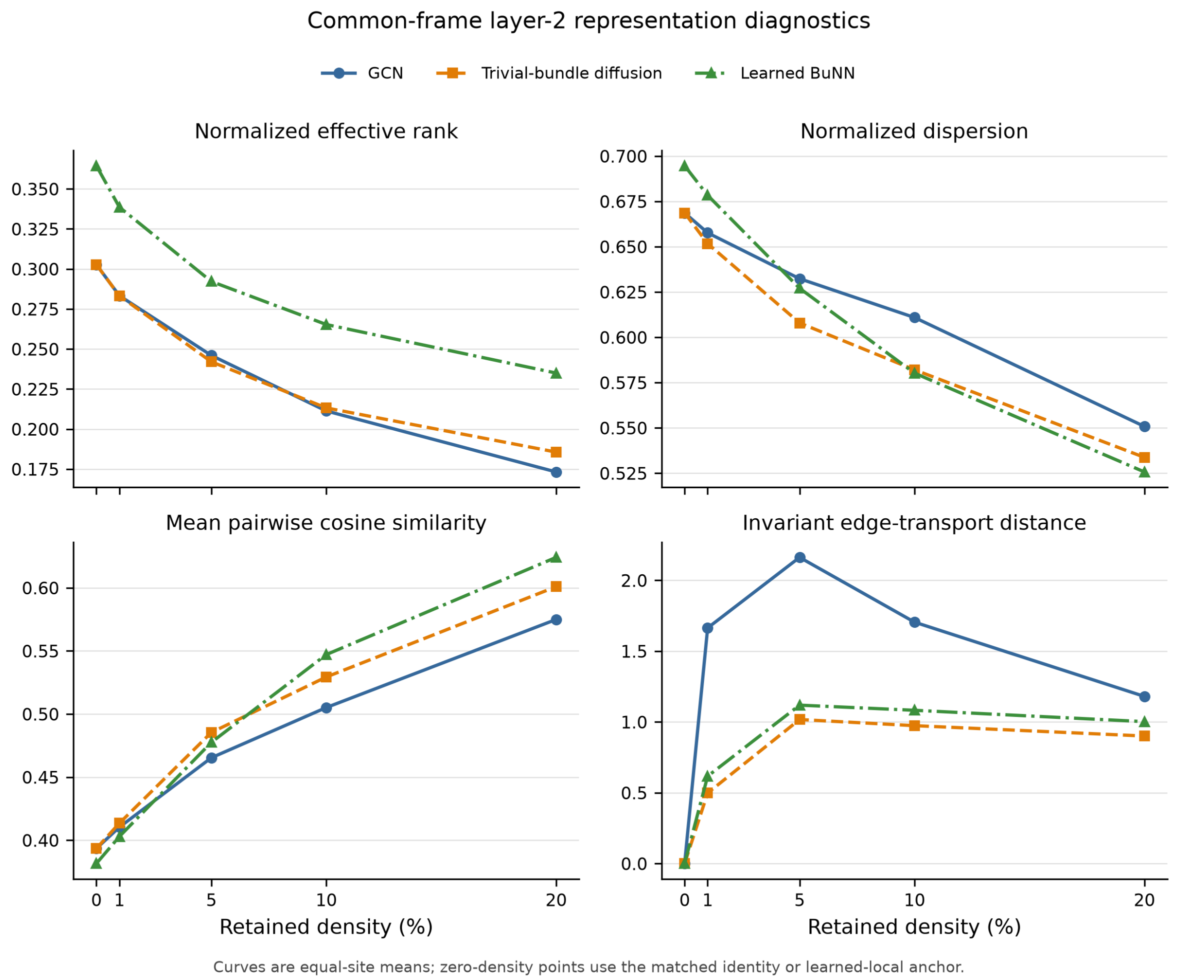

Figure 4: Equal-site common-frame representation diagnostics at layer 2. Zero-density points use the matched identity or learned-local anchor. Raw, unaligned BuNN node embeddings were not compared.

4.4 Site, seed, and weighting sensitivity

Only 1 of the 18 leave-one-site influence estimates was positive, and no favorable exclusion interval was entirely above zero. Two of five seed point estimates favored BuNN, but neither favorable interval excluded zero; one unfavorable interval was entirely below zero. Participant weighting reversed the sign of the small BuNN versus GCN point estimate to +0.0030 (95% site-cluster bootstrap interval $[-0.0077, 0.0155]$), whereas the equal-site mean and median remained negative. The pre-specified confirmatory estimand was the equal-site mean.

Table 9: Sensitivity of the three core contrasts to site weighting.

Contrast	**Equal-site mean**	**Participant-weighted**	**Site median**
BuNN curve − GCN curve	-0.0096	+0.0030	-0.0044
BuNN curve − elastic net	-0.0552	-0.0445	-0.0464
BuNN − GCN effective-rank change	-0.0072	-0.0109	-0.0092

Table 10: Post-hoc model summaries under equal-site and participant weighting. Intervals resample held-out sites as clusters.

Model	Equal-site mean [95% CI]	Participant-weighted mean [95% CI]
Connectome elastic net	0.640 [0.601, 0.678]	0.647 [0.617, 0.681]
RBF-SVM	0.626 [0.578, 0.673]	0.656 [0.618, 0.683]
Identity	0.603 [0.556, 0.643]	0.618 [0.581, 0.645]
GCN curve	0.595 [0.552, 0.635]	0.600 [0.568, 0.627]
Learned-local	0.605 [0.561, 0.645]	0.622 [0.593, 0.646]
Learned BuNN curve	0.585 [0.545, 0.621]	0.603 [0.570, 0.632]

Table 11: Post-hoc weighting sensitivity for the main practical contrasts.

Contrast	Equal-site estimate [95% CI]	Participant-weighted estimate [95% CI]
BuNN curve − GCN curve	-0.010 [-0.036, +0.011]	+0.003 [-0.008, +0.015]
BuNN curve − elastic net	-0.055 [-0.083, -0.027]	-0.044 [-0.065, -0.022]
RBF-SVM − elastic net	-0.015 [-0.055, +0.020]	+0.009 [-0.031, +0.035]

Replacing the trapezoid with a uniform average across the five density points changed the BuNN-minus-GCN estimate from -0.0096 to -0.0066 (95% interval $[-0.0269, 0.0113]$). The result therefore did not depend on the trapezoid's heavier weighting of the 10 and 20 percent graphs.

4.5 Heat-operator regime

The spectral audit showed that the graph regime changed sharply with density. At 1 percent density, a mean 60.5 percent of Laplacian modes were stationary, which reflects many disconnected components and isolated nodes. This fraction fell to 16.5, 7.8, and 3.3 percent at 5, 10, and 20 percent density. At the three denser settings, the median eigenvalue was close to one, so a typical nonstationary mode retained about $e^{-1} = 0.368$ of its amplitude in one heat step. The fixed $t = 1$ operator was therefore not uniformly mild across the density grid.

Table 12: Post-hoc spectrum of the implemented heat operator, averaged over 754 participant graphs at each density. A stationary mode has $\lambda = 0$ and heat response one.

Density	λ_2	Median λ	Mean $e^{-\lambda}$	Stationary-mode fraction
1%	0.000	0.001	0.739	0.605
5%	0.000	0.990	0.492	0.165
10%	0.003	1.037	0.430	0.078
20%	0.056	1.032	0.394	0.033

4.6 Post-hoc nonlinear connectome baseline

The RBF-SVM reached 0.6255 equal-site balanced accuracy, 0.6565 pooled balanced accuracy, and 0.7014 pooled AUROC. Against the connectome elastic net, its equal-site difference was -0.0146 (95% paired site-bootstrap interval $[-0.0555, 0.0206]$; exact sign-flip $p = 0.498$). The analysis therefore found no RBF-SVM advantage over the strongest classical baseline.

RBF-SVM exceeded the GCN curve by 0.0310, but that interval crossed zero. Its difference from the learned-BuNN curve was 0.0406 (95% interval $[0.0069, 0.0744]$; exact $p = 0.037$). This is a post-hoc practical comparison, not a new confirmatory test of the BuNN hypothesis. The three RBF comparisons were not assigned a confirmatory multiplicity family. In a descriptive Holm sensitivity across those three tests, the adjusted values were 0.111 for RBF-SVM versus BuNN, 0.386 versus GCN, and 0.498 versus elastic net. The unadjusted $p = 0.037$ is therefore not treated as a standalone discovery.

Weighting changed the model ordering. RBF-SVM was 0.0146 below elastic net in the equal-site mean but 0.0089 above it when sites were weighted by participant count. The median paired site difference was zero. Pooled performance alone would therefore have suggested a different conclusion from the pre-specified equal-site analysis.

Table 13: Post-hoc RBF-SVM comparisons using paired held-out sites.

Contrast	Difference	95% CI	Exact p
RBF-SVM − elastic net	-0.0146	[−0.0555, +0.0206]	0.498
RBF-SVM − GCN curve	+0.0310	[−0.0156, +0.0705]	0.193
RBF-SVM − learned BuNN curve	+0.0406	[+0.0069, +0.0744]	0.037

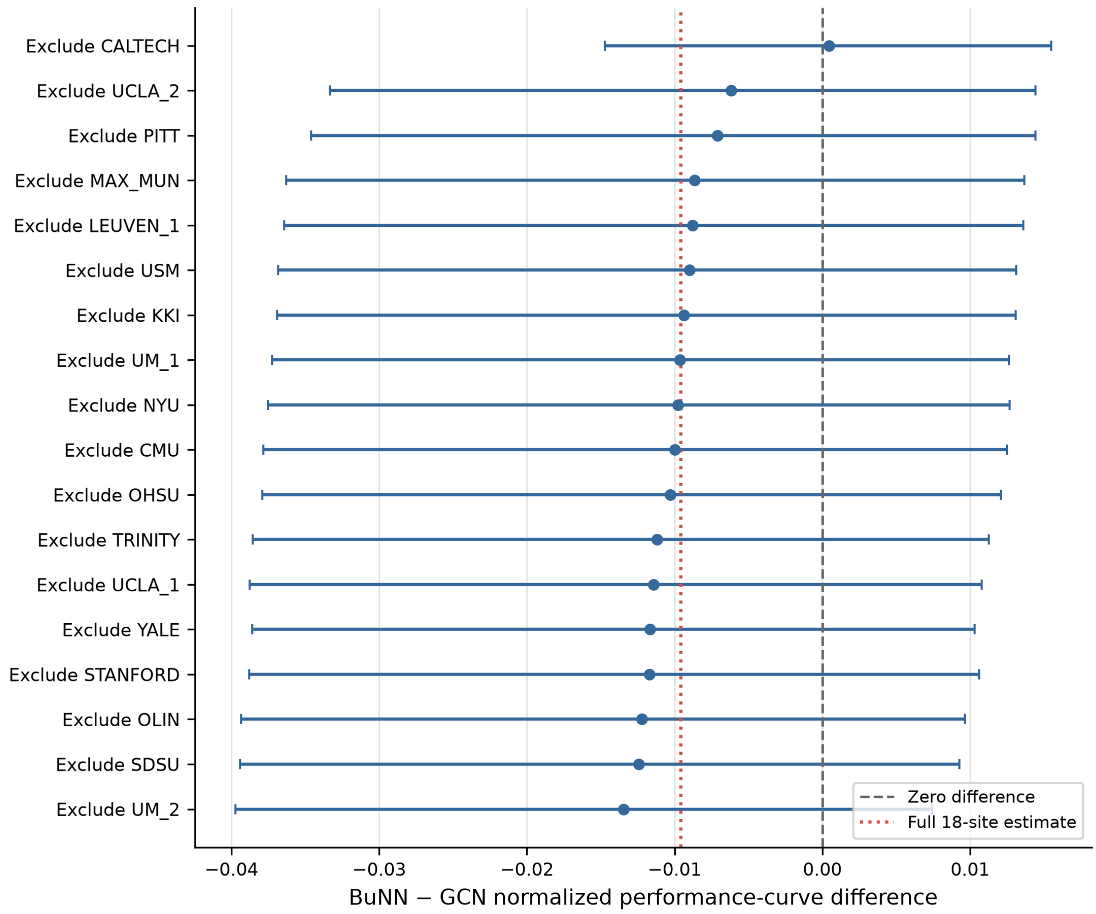

Figure 5: Leave-one-site influence on the BuNN versus GCN curve contrast. This analysis checks whether a single site drives the confirmatory average; it does not turn the 18 sites into independent replications.

4.7 Execution cost

Learned BuNN used 8,529 parameters, compared with 5,889 for GCN. The recorded BuNN fits required 31.14 hours in total, while GCN required 15.09. Peak GPU memory stayed below 0.38 GiB for both. These models and graphs are small, so the workload consisted of many independent fits rather than one large tensor. Runtime values apply only to this implementation and the A10G system.

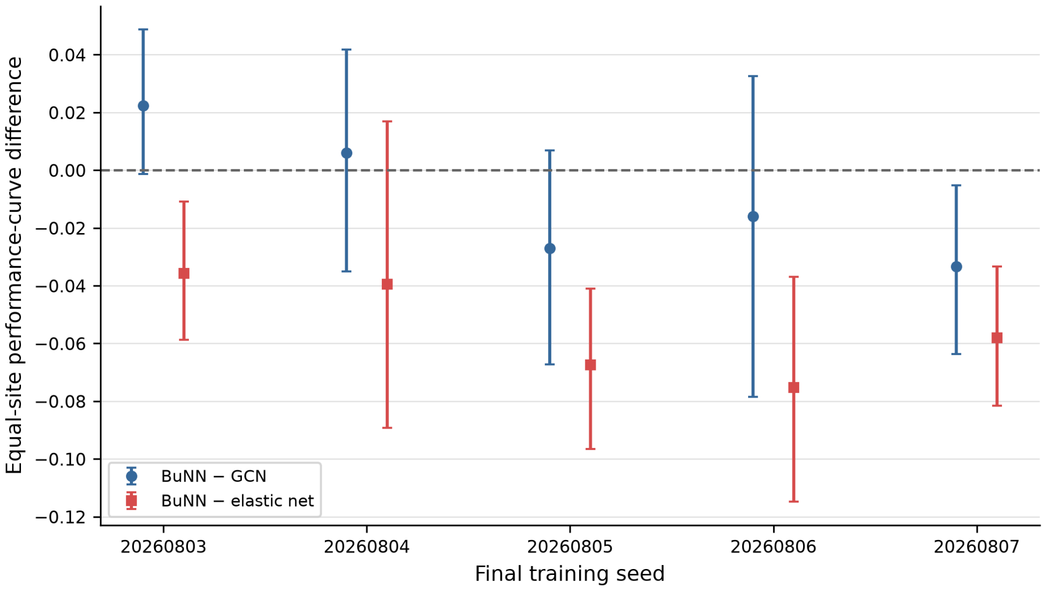

Figure 6: BuNN versus GCN contrast across the five fixed final training seeds. The variation shows that the small average difference is sensitive to optimization randomness.

Table 14: Observed operator cost and equal-site curve performance.

Operator	Parameters	Fit-hours	Peak GPU GiB	Curve BA
GCN	5,889	15.09	0.373	0.5945
Trivial bundle	5,889	25.99	0.377	0.5911
Learned BuNN	8,529	31.14	0.378	0.5849

4.8 E1: accepted-checkpoint intervention results

The learned maps were not dormant. Replacing them with random orthogonal maps reduced held-out balanced accuracy by 8.86 percentage points (95% paired site-bootstrap interval $[-12.36, -4.78]$; exact sign-flip $p = 0.00095$; Holm-adjusted $p = 0.0038$). The effect was negative at every density and in 15 of 18 sites after densities were averaged. Random-map replacement was the only intervention to pass the pre-specified multiplicity-controlled test.

Identity maps reduced balanced accuracy by 4.72 points (95% interval $[-8.56, -0.29]$). The unadjusted exact p was 0.0468, but the adjusted value was 0.1404. Shuffling learned maps between nodes produced a 1.94-point reduction (95% interval $[-3.77, 0.05]$; adjusted $p = 0.1404$), and degree-preserving rewiring produced a 1.23-point reduction (95% interval $[-2.49, -0.05]$; adjusted $p = 0.1404$). Bootstrap intervals and sign-flip tests answer different statistical questions. The adjusted sign-flip result controls the four named primary tests.

The revision analysis also compared the interventions directly. Random maps caused 6.92 points more damage than shuffled learned maps and 7.63 points more than topology rewiring;

Table 15: Accepted-checkpoint intervention effects. Estimates are intervened minus unaltered BuNN balanced accuracy in percentage points.

Intervention	**Estimate**	**95% CI**	**Holm p**
Identity maps	-4.72	$[-8.56, -0.29]$	0.1404
Shuffled learned maps	-1.94	$[-3.77, 0.05]$	0.1404
Random orthogonal maps	-8.86	$[-12.36, -4.78]$	**0.0038**
Degree-preserving rewiring	-1.23	$[-2.49, -0.05]$	0.1404

both six-comparison Holm-adjusted values were 0.009. The other four pairwise comparisons did not survive that correction. This supports a limited statement: an arbitrary replacement of the checkpoint's coordinate system was more disruptive than reassigning the learned map collection or rewiring the selected graph. It does not identify a biological node geometry.

Table 16: Post-hoc pairwise E1 intervention comparisons. Positive values mean that the first intervention caused a less negative change in balanced accuracy.

Exploratory E1 contrast	Difference (pp)	95% CI	Holm p
Identity maps − shuffled learned maps	-2.78	[-5.25, +0.43]	0.253
Identity maps − random orthogonal maps	+4.14	[+0.59, +7.95]	0.180
Identity maps − topology rewiring	-3.49	[-7.22, +0.81]	0.253
Shuffled learned maps − random orthogonal maps	+6.92	[+3.43, +10.02]	0.009
Shuffled learned maps − topology rewiring	-0.71	[-2.61, +1.71]	0.586
Random orthogonal maps − topology rewiring	-7.63	[-10.98, -3.88]	0.009

The random-map effect remained negative at 1, 5, 10, and 20 percent density (figure 8). Identity maps had their largest effect at 1 percent. Map shuffling and topology rewiring produced smaller effects, often with density-specific intervals that crossed zero. The random-map result was thus not confined to one selected density.

Random maps changed 49.46% of binary decisions and reduced AUROC by 14.59 points. Identity maps changed 34.41% of decisions and reduced AUROC by 9.50 points. The corresponding decision-change rates were 15.12% for shuffled maps and 6.96% for topology rewiring. Random maps reduced sensitivity while increasing specificity, a shift toward control predictions rather than symmetric noise.

Every intervention changed at least one gauge-aware diagnostic, but those changes tracked predictive effects poorly. Across 72 site-density cells, the absolute descriptive Spearman correlations between diagnostic changes and balanced-accuracy changes never exceeded 0.21 (figure 9). The available summaries do not reveal a simple predictive mechanism and cannot support a causal mediation claim.

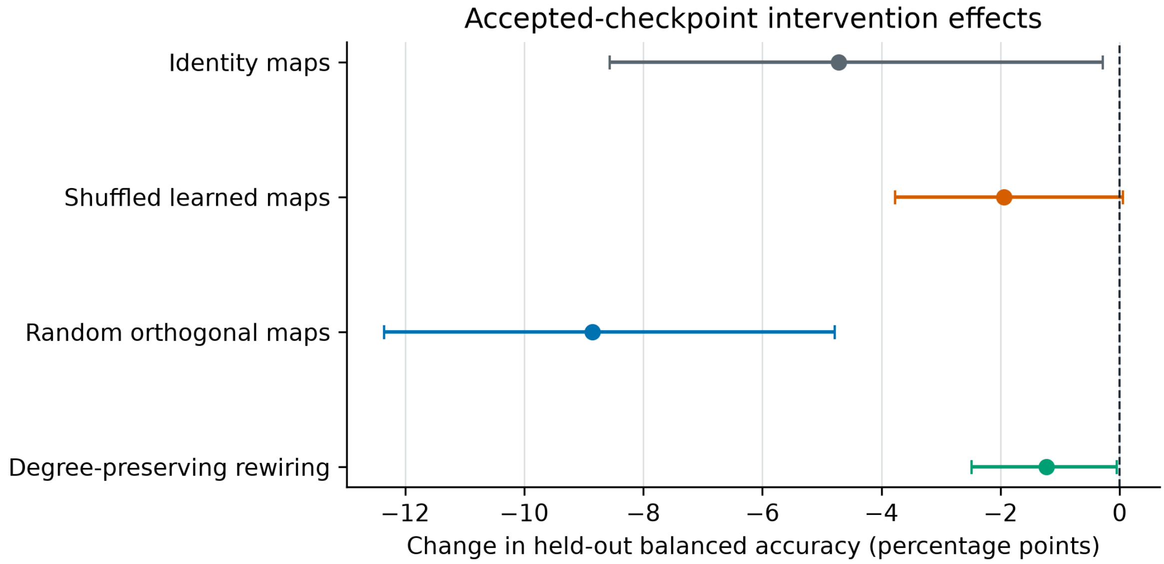

Figure 7: Overall E1 intervention effects with 95% paired site-bootstrap intervals. Negative values mean that disrupting the component reduced the accepted checkpoint's held-out balanced accuracy. Statistical decisions use the Holm-adjusted exact sign-flip tests in table 15.

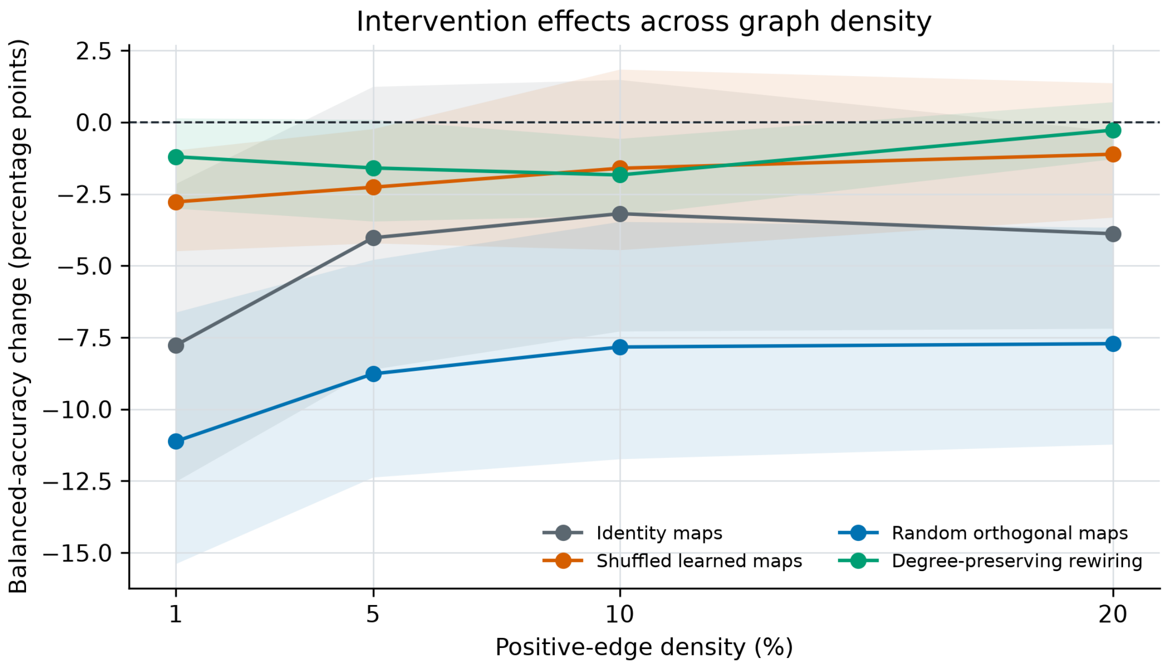

Figure 8: Density-specific accepted-checkpoint intervention effects. Shaded regions are 95% paired site-bootstrap intervals. These density-specific estimates are descriptive; the primary inference averages the four effects within site before equal-site analysis.

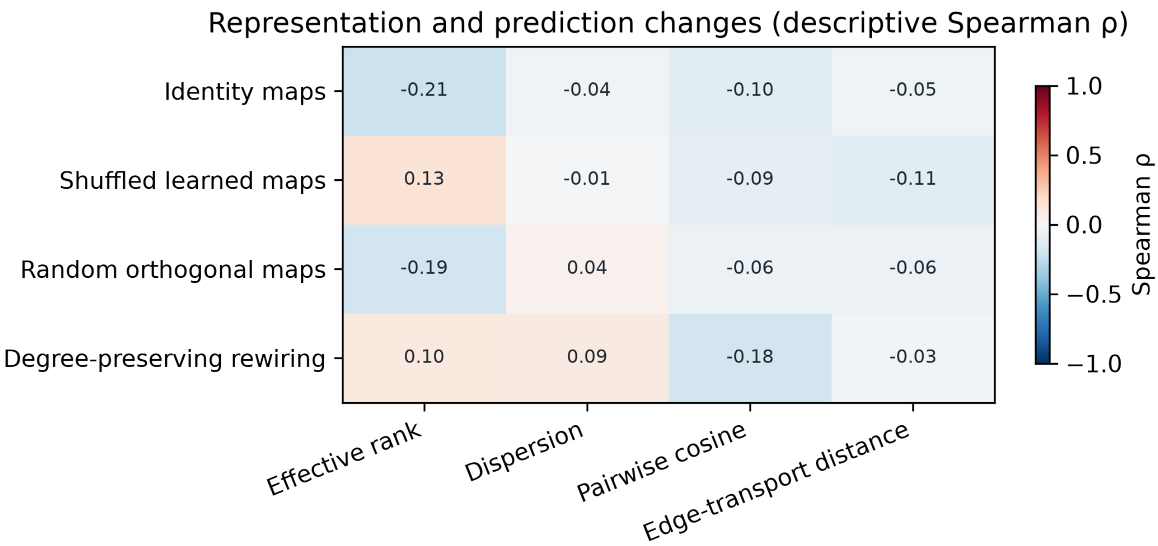

Figure 9: Descriptive association between final-layer gauge-aware diagnostic changes and balanced-accuracy changes across 72 site-density cells. The correlations are exploratory summaries, not multiplicity-controlled tests or evidence of causal mediation.

4.9 E2: synthetic known-geometry results

E2 met its frozen support rule. The conditional learned-BuNN advantage was 42.75 percentage points (95% paired bootstrap interval $[38.42, 47.67]$; exact sign-flip $p = 0.001953$). This contrast measures the increase in the BuNN versus GCN difference from S0 to S1, paired across ten replicate seeds. It exceeded the three-point practical threshold, and the interval excluded zero. The exact value 0.001953 is the smallest possible two-sided sign-flip value for ten pairs; all ten paired contrasts had the same sign.

Table 17: Pre-specified E2 validity gates. Differences are test balanced accuracy in percentage points.

Frozen contrast	**Estimate**	**95% CI**	**Exact p**
Conditional BuNN advantage	42.75	$[38.42, 47.67]$	0.001953
Oracle − GCN in S1	43.50	$[39.42, 48.00]$	0.001953
BuNN − learned-local in S1	48.67	$[45.83, 51.50]$	0.001953
BuNN − GCN in S0	−0.25	$[-0.58, 0.00]$	0.500000

The oracle positive control and matched-capacity gate passed, and the S0 capacity warning did not trigger (table 17 and figure 10). That warning had limited power because both models were near the ceiling in S0. In S0, GCN and learned BuNN reached balanced accuracies of 1.000 and 0.998, showing that ordinary aggregation solved the task when frames were aligned. In S1, learned BuNN reached 0.990 and the oracle reached 1.000, compared with 0.565 for GCN and 0.503 for learned-local.

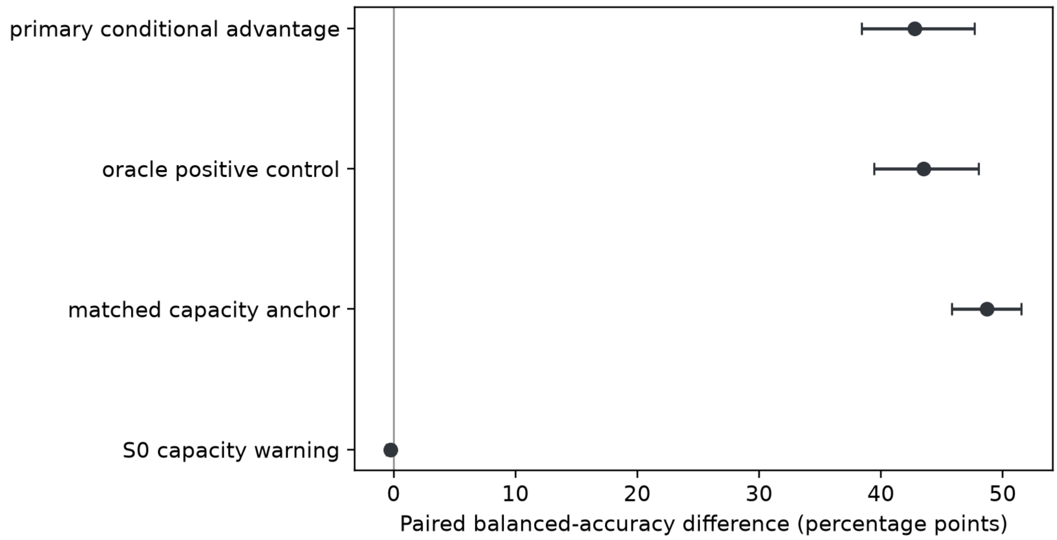

Figure 10: E2 paired validity contrasts with 95% bootstrap intervals across ten replicate seeds. The first three contrasts are ordered support gates; the S0 contrast checks for a generic BuNN advantage when alignment is not required.

The corruption controls confined that success to the intended conditions (figure 11). Wrong topology reduced learned-BuNN balanced accuracy to 0.493 and the oracle to 0.517. Shuffling the label-independent frame marker reduced learned BuNN to 0.503, while the oracle still received the true signal maps and remained at 1.000. When frames varied independently between participants without an informative marker, learned BuNN reached 0.498 and the oracle remained at 1.000. Relative to S1, the paired reductions for wrong topology, shuffled markers, and unlearnable frames were 49.67, 48.67, and 49.17 points; every exact p was 0.001953.

The marker-noise sweep produced a graded response (figure 12). Mean learned-BuNN balanced accuracy was 0.990 at zero degrees, 0.992 at 15 degrees, 0.990 at 30 degrees, 0.891 at 60 degrees, and 0.511 at 120 degrees. Variation between seeds was large at 60 degrees (SD 0.202). The curve locates a boundary in this experiment but does not define a general angular tolerance.

S6 copied a whole-graph statistic to every node. All seven operators reached 1.000 balanced accuracy. In this controlled case, message passing became redundant because each node already disclosed the relevant global information.

Learned BuNN did not recover the planted maps exactly. Its mean gauge-invariant relative-transport error in S1 was 22.07, compared with 30.62 for identity-frame operators and zero for the oracle. The learned transport was useful for the task, but near-perfect classification did not imply exact map identification.

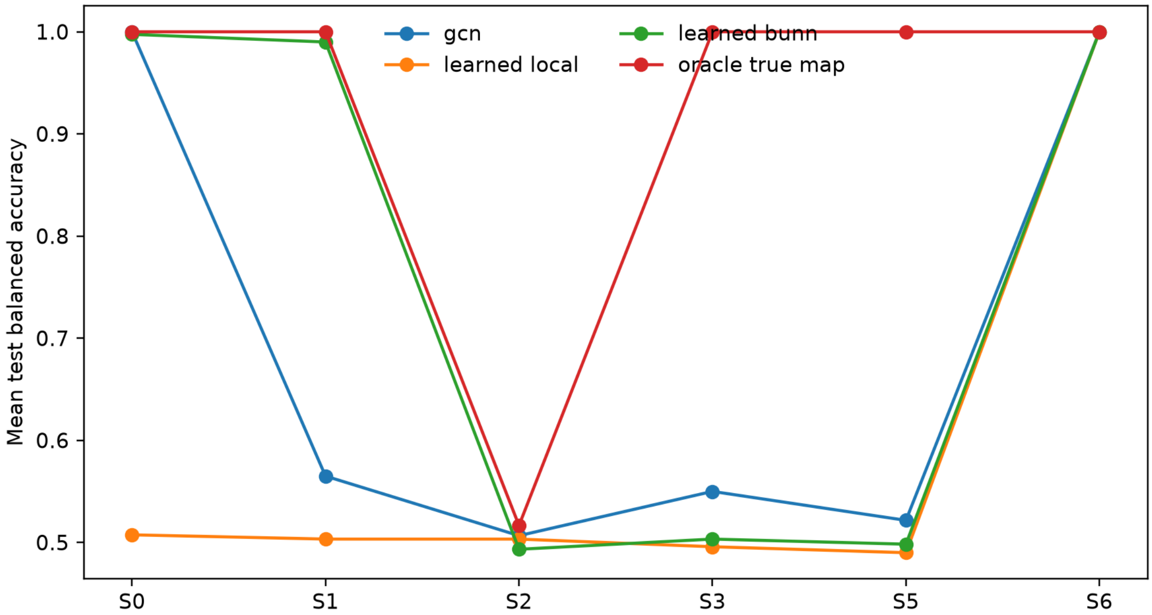

Figure 11: Mean test balanced accuracy across E2 conditions. S1 provides fixed recoverable geometry; S2, S3, and S5 break topology, marker assignment, and across-participant learnability. S6 supplies an already-global summary.

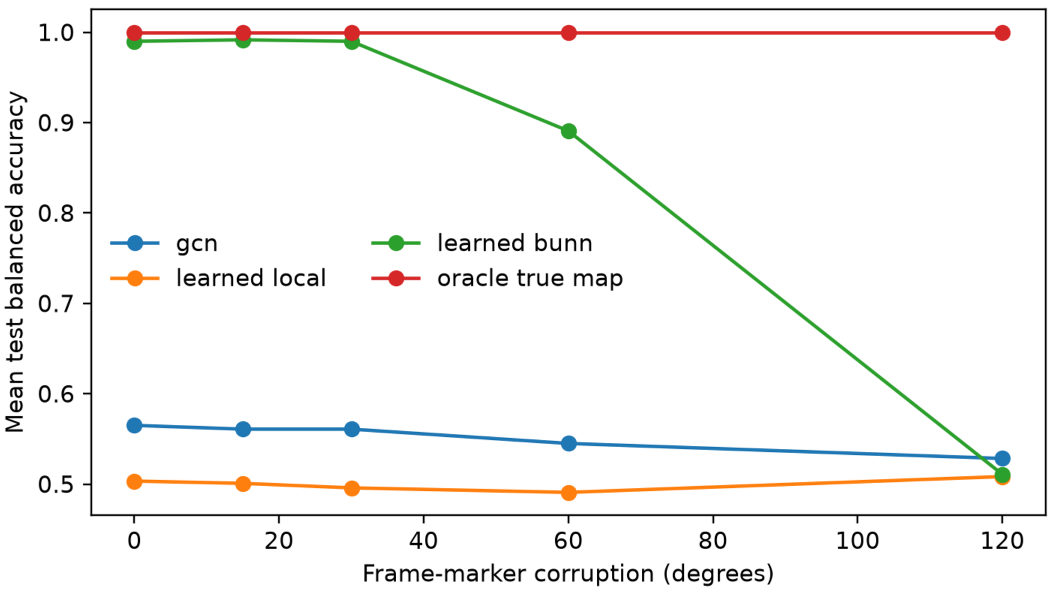

Figure 12: Performance across corruption of the frame marker. The oracle continues to receive true signal maps; learned BuNN must infer a useful transport from progressively less reliable marker information.

5 Discussion

The proposed advantage did not appear in the ABIDE-I experiment. Learned bundle transport neither improved the performance curve over GCN nor exceeded the regularized connectome baseline. Its matched-anchor effective rank was not better preserved. Because the inputs, backbone, tuning budget, and controls were shared, the comparison isolates this propagation design rather than contrasting unrelated models.

The later RBF-SVM comparison reached the same practical conclusion from a different model family. Its point estimate was above learned BuNN, but the comparison was post-hoc and did not remain below 0.05 in the three-contrast Holm sensitivity. It did not improve on elastic net under equal-site weighting. The strongest equal-site reference in this pipeline therefore remained the simpler regularized linear model.

E1 argues against a completely dormant map system. Random orthogonal replacement caused a large loss of held-out performance that survived the four-intervention correction, and it caused more damage than map shuffling or topology rewiring in the six-pair revision analysis. This intervention also breaks the coordinate system expected by later weights. It shows checkpoint dependence, not that the maps recovered node-specific brain geometry. The transformations were consequential without producing better cross-site prediction than GCN or the elastic net.

E2 shows how both findings can be true. Learned BuNN clearly exceeded GCN and learned-local when the graph was correct and the coordinate frames were fixed and recoverable. Its advantage disappeared when the topology was wrong, the frame marker was shuffled, or the frames changed between participants without recoverable information. The operator can solve a task that requires bundle transport without improving the selected ABIDE-I representation.

5.1 Why the result was plausible before the experiment

The original hypothesis followed directly from the operator. Ordinary diffusion can contract node representations, whereas orthogonal transport can align neighboring features before they are averaged. When those features use different but informative frames, alignment can retain structure that direct averaging removes. BuNN theory and earlier synthetic experiments gave this idea a reasonable basis [3].

Functional connectomes pose a harder test. They are dense, and every node feature is already a global profile. Transporting local information may offer little benefit when each node already describes its relationship with every other node.

5.2 What the completed evidence can explain

No single mechanism is established. The observations instead constrain which explanations remain plausible.

The most consistent explanation concerns the architecture and its inputs. Full connectivity rows already expose global information, while denser propagation averages those rows. Learned orthogonal maps altered BuNN's predictions, but degree-preserving graph rewiring had a much smaller measured effect. The model could therefore learn useful internal transformations without gaining from message passing over the selected positive-edge graph. Those transformations also failed to exceed a GCN or a linear model that directly used the complete connectome.

The spectral audit makes the density effect more concrete. At 1 percent, the graphs contained many disconnected components, so a large part of the signal lay in stationary modes. At 5 to 20 percent, the heat operator retained only about 0.35 to 0.37 of a median nonstationary mode in each layer. Two layers can therefore impose substantial contraction even though the networks are shallow. This is a property of the implemented operator and graph rule, not proof that over-smoothing caused the classification result. Learned transport may align some fields before this contraction, but the matched-anchor diagnostics and held-out predictions show no measured advantage from that alignment here.

E2 separates two conditions behind this interpretation. Learned BuNN succeeded when the graph encoded the relevant smoothness relation and the frame information was recoverable. It failed when either condition was broken. Once a global summary was copied to every node, all operators succeeded without a propagation advantage. Study 1, E1, and E2 are consistent with this account, but they do not show that the same components or coordinate frames exist in fMRI.

5.3 Why the learned-local control matters

BuNN adds parameters through its map generator. If GCN were the only control, differences caused by that capacity could be misattributed to transport. The learned-local model keeps the generator but removes diffusion between nodes. Its high zero-density common-frame effective rank shows why absolute aligned embeddings alone would overstate preservation. Raw unaligned BuNN embeddings were never used. Each operator must instead be compared with its own no-propagation state.

Table 18: Evidence-based interpretation of the null result.

Observation	**Supported interpretation**	**What it does not prove**
Diffusion curves decline with density	Adding more propagation edges is associated with worse held-out-site prediction in this pipeline.	A universal law of over-smoothing or a causal biological mechanism.
BuNN does not beat GCN	Learned transport provides no detected predictive advantage over ordinary aggregation here.	That BuNN never works on brain or non-brain graphs.
BuNN remains below elastic net	A regularized model on the full connectome is the stronger practical reference for these inputs.	That nonlinear modeling is always unnecessary.
RBF-SVM does not beat elastic net	A nonlinear whole-connectome model did not improve the equal-site result.	That every nonlinear model or feature representation will fail.
Learned-local anchor already has high common-frame rank	Extra node-wise map capacity can raise absolute aligned diversity without transport.	That every aspect of the learned maps is useless.
Matched-anchor rank favors GCN	The proposed common-frame representation-preservation advantage is not supported.	A causal mediation pathway from rank to prediction.
Random maps reduce BA by 8.86 points	Accepted predictions depend on properties of the learned map system.	Biological bundle geometry or causal neural transport.
Random replacement is worse than map shuffling	The accepted coordinate system matters more than node reassignment under these interventions.	That the learned maps identify brain-specific coordinate frames.
Degree-preserving rewiring reduces BA by 1.23 points	The constructed propagation graph contributes modestly to prediction.	Anatomical validity of the graph or a universal topology effect.
Heat spectra change with density	One-percent graphs are highly disconnected; denser graphs strongly attenuate typical nonstationary modes at $t = 1$.	That spectral contraction caused the predictive decline.
Diagnostic and performance correlations are weak	Large representation changes do not reliably co-vary with predictive changes across site-density cells.	Proof that representations can never mediate prediction.
E2 BuNN succeeds only in S1	Learned transport helps when topology dependence and fixed recoverable frames are deliberately present.	That ABIDE-I has the same planted geometry.
E2 controls fall to chance	Correct topology and recoverable frame information are necessary for this synthetic advantage.	A universal failure threshold for real connectomes.
Every S6 operator is perfect	Already-global inputs can remove the need for propagation in a controlled setting.	That ABIDE connectivity rows equal the synthetic summary.
Site and seed sensitivity	The small BuNN versus GCN difference is not stable to all reasonable weighting and optimization choices.	That the full result is invalid or that site effects explain every difference.

5.4 Site heterogeneity and uncertainty

The primary BuNN versus GCN interval crossed zero, and participant weighting reversed the sign of the point estimate. Excluding individual sites or changing the training seed altered the size and sometimes the direction of this small difference. The models should not be called equivalent. The supported statement is narrower: no BuNN advantage was detected. The equal-site upper bound was 0.0115, below the protocol's 0.03 reference margin, but that margin does not support a formal equivalence claim.

The elastic-net comparison was more stable. Its interval remained below zero, so learned BuNN underperformed the regularized non-graph reference in this analysis. The post-hoc RBF-SVM result depended on weighting: elastic net led under equal-site averaging, whereas RBF-SVM led under participant weighting. A comparison limited to neural graph variants or pooled scores would have missed these distinctions.

5.5 Limitations

The conclusions apply to ABIDE-I, C-PAC filtered data without global-signal regression, the AAL-116 atlas, Pearson correlations with Fisher transformation, positive-edge propagation graphs, full connectivity-row node features, the 0 to 20 percent density grid, the shared two-layer backbone, and the fixed tuning budget. Other atlases, preprocessing pipelines, graph rules, node features, or BuNN architectures may behave differently.

Functional correlations are observational. They do not identify anatomy, excitation, inhibition, or causal neural communication. Technical quality control also does not remove demographic or clinical sampling bias. Site sizes differ, so equal-site and participant-weighted means answer different questions. The representation diagnostics are computational summaries, and their association with prediction is not a mediation analysis.

E1, E2, and the RBF-SVM comparator were specified after the Study 1 result, so they are post-hoc analyses rather than independent confirmation. E1's interventions are intentionally artificial. Random maps can disrupt the coordinate system expected by later weights. Identity maps remove learned orientation and reflections. Node shuffling preserves the collection of learned maps while breaking their assignment. The measured effects show dependence within the accepted checkpoints, not the effect of changing one isolated biological mechanism.

The report also lacks an external ABIDE-II model evaluation. A score-blind availability audit found the official phenotype and quality records for 1,114 participants but no complete C-PAC filtered, no-GSR AAL-116 ROI-time-series derivative. The alternative LLE repository documents a mixture of C-PAC and SPM-preprocessed volumes rather than the frozen Study 1 representation [15, 16]. The compatibility gate therefore failed before any model was fitted or scored.

E2 was also designed after the ABIDE-I result. Its ring graph, frame marker, smooth signal, noise scale, and global summary are controlled objects, not estimates of brain organization. Near-perfect S0, S1, oracle, and S6 scores show that the positive controls were strong, but say nothing about less separable synthetic families. Learned BuNN did not recover the true relative transports exactly. E2 supports task-relevant alignment, not exact geometric identification.

The stopped version 1 E2 run exposed an error before unblinding: zero angles still produced reflections in half of the bundles. That run was quarantined, an explicit identity regression test was added, and version 2 was trained from scratch. The incident shows why mechanism studies need generator-level tests.

5.6 Unfinished extensions

E1 and E2 are complete and independently audited, but their protocols were frozen after Study 1 was known. They remain separate from the confirmatory result. Matched deeper GCN and BuNN networks and a leakage-free Wang-style time-series model would be new studies. ABIDE-II was examined only for metadata and derivative availability. External evaluation remains blocked unless a matching official derivative appears or a separate raw-data preprocessing protocol is frozen and validated. None of these extensions is required to state the current result.

6 Conclusion

In the pre-specified ABIDE-I analysis, learned orthogonal bundle transport did not improve the held-out-site performance curve over GCN. No advantage was detected over identity propagation or the learned-local control because both intervals included zero. BuNN was below the regularized connectome baseline, whose paired interval was entirely negative. Common-frame diagnostics, measured relative to matched anchors, did not support stronger representation preservation.

This result is limited to the tested design. The checkpoint audit showed that the maps were active: random orthogonal replacement reduced balanced accuracy by 8.86 points after multiplicity correction. Exact node-map assignment and degree-preserving topology had smaller measured effects, and changes in the representation summaries were only weakly associated with changes in prediction.

E2 established a setting in which transport was useful. Learned BuNN gained 42.75 points on the conditional contrast and passed the oracle and learned-local gates. That advantage disappeared when topology was wrong, frame information was shuffled or unrecoverable, or each node already received the relevant global feature. Bundle transport can therefore perform its intended computation, but the selected ABIDE-I graph and globally informed node features did not turn that computation into better cross-site classification. This is a conditional computational result, not evidence for or against biological bundle geometry.

A Evidence boundary and remaining work

Table 19: Boundary between completed evidence and planned work.

Component	**Completed and audited**	**Not yet a result**
Data	ABIDE-I C-PAC filtered, no-GSR AAL time series; technical QC; 754-person connectome archive; score-blind ABIDE-II metadata and derivative-availability audit.	Alternative preprocessing, atlases, negative-edge propagation, and any ABIDE-II model evaluation.
Classical models	Covariate, connectome elastic-net, and combined baselines across all 18 held-out sites.	Additional harmonization or feature-selection families.
Neural models	Two-layer identity, learned-local, GCN, trivial-bundle, and learned-BuNN operator study; five seeds; complete density grid.	Five-layer matched study, Wang-inspired time-series architecture.
Mechanism evidence	Common-frame representation curves, matched anchors, site influence, seed stability, efficiency, the audited 360-checkpoint intervention study, and the 700-cell synthetic known-geometry study.	Optional matched depth/capacity sensitivity and component-level feature-loss analysis.
External validity	None claimed beyond the frozen ABIDE-I pipeline; the ABIDE-II compatibility gate failed before prediction.	Prospective ABIDE-II evaluation only after a matching derivative or separately frozen raw-data pipeline passes the gate.

B Reproducibility record

The repository separates accepted scientific artifacts from presentation files. Raw participant derivatives, checkpoints, and credentials are excluded from the public release. Numbers in the report come from generated tables and macros tied to accepted analysis hashes. Before Study 1 was interpreted, the following checks had passed:

- all 18 outer sites were present;
- expected tuning, prediction, metric, diagnostic, and runtime row counts were complete;
- model and data contracts matched their frozen hashes;
- hidden metrics were independently recomputed before unblinding;
- parallel worker outputs matched the validated sequential execution in the engineering equivalence test;
- site-bootstrap and sign-flip analyses were built and tested on synthetic data before use on accepted results;
- all 700 accepted E2 cells passed score-blind cell audits and all 100 condition-replicate groups were independently sealed before unblinding;
- the E2 analysis was invoked once under its frozen acknowledgment and independently recomputed artifact by artifact;
- report figures and tables were regenerated from the frozen result snapshot rather than edited manually.

The project repository contains the source code, report source, manifests, and AI-use log. This PDF contains no participant-level source data.

References

[1] K. Han, Y. Su, L. He, L. Zhan, S. Plis, V. Calhoun, and C. Yang, "Rethinking functional brain connectome analysis: Do graph deep learning models help?" *npj Artificial Intelligence*, vol. 2, p. 19, 2026. [Online]. Available: https://doi.org/10.1038/s44387-025-00067-x

[2] A. Abraham, M. P. Milham, A. Di Martino, R. C. Craddock, D. Samaras, B. Thirion, and G. Varoquaux, "Deriving reproducible biomarkers from multi-site resting-state data: An autism-based example," *NeuroImage*, vol. 147, pp. 736–745, 2017. [Online]. Available: https://doi.org/10.1016/j.neuroimage.2016.10.045

[3] J. Bamberger, F. Barbero, X. Dong, and M. M. Bronstein, "Bundle neural network for message diffusion on graphs," in *International Conference on Learning Representations*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/2ac879d1865475a7abc8dfc7a9c15c27-Abstract-Conference.html

[4] C. Bodnar, F. Di Giovanni, B. P. Chamberlain, P. Li‘o, and M. M. Bronstein, “Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2022/hash/75c45fca2aa416ada062b26cc4fb7641-Abstract.html

[5] K. Murphy and M. D. Fox, “Towards a consensus regarding global signal regression for resting state functional connectivity MRI,” *NeuroImage*, vol. 154, pp. 169–173, 2017. [Online]. Available: https://doi.org/10.1016/j.neuroimage.2016.11.052

[6] A. Di Martino *et al.*, “The autism brain imaging data exchange: Towards a large-scale evaluation of the intrinsic brain architecture in autism,” *Molecular Psychiatry*, vol. 19, no. 6, pp. 659–667, 2014. [Online]. Available: https://doi.org/10.1038/mp.2013.78

[7] Preprocessed Connectomes Project, “ABIDE preprocessed,” INDI, Child Mind Institute, 2026, accessed 9 August 2026. [Online]. Available: https://fcon_1000.projects.nitrc.org/indi/abide/preprocessed.html

[8] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *International Conference on Learning Representations*, 2017. [Online]. Available: https://openreview.net/forum?id=SJU4ayYgl

[9] A. Roth and T. Liebig, “Rank collapse causes over-smoothing and over-correlation in graph neural networks,” in *Proceedings of the Second Learning on Graphs Conference*, ser. Proceedings of Machine Learning Research, vol. 231, 2024, pp. 35:1–35:23. [Online]. Available: https://proceedings.mlr.press/v231/roth24a.html

[10] S. Parisot, S. I. Ktena, E. Ferrante, M. C. H. Lee, R. Guerrero, B. Glocker, and D. Rueckert, “Disease prediction using graph convolutional networks: Application to autism spectrum disorder and alzheimer’s disease,” *Medical Image Analysis*, vol. 48, pp. 117–130, 2018. [Online]. Available: https://doi.org/10.1016/j.media.2018.06.001

[11] L. Wang, K. Li, and X. P. Hu, “Graph convolutional network for fMRI analysis based on connectivity neighborhood,” *Network Neuroscience*, vol. 5, no. 1, pp. 83–95, 2021. [Online]. Available: https://doi.org/10.1162/netn_a_00171

[12] T. Yang, M. A. Al-Duailij, S. Bozdag, and F. Saeed, “Classification of autism spectrum disorder using rs-fMRI data and graph convolutional networks,” in *2022 IEEE International Conference on Big Data*, 2022, pp. 3131–3138. [Online]. Available: https://doi.org/10.1109/BigData55660.2022.10021070

[13] N. Tzourio-Mazoyer, B. Landeau, D. Papathanassiou, F. Crivello, O. Etard, N. Delcroix, B. Mazoyer, and M. Joliot, “Automated anatomical labeling of activations in SPM using a macroscopic anatomical parcellation of the MNI MRI single-subject brain,” *NeuroImage*, vol. 15, no. 1, pp. 273–289, 2002. [Online]. Available: https://doi.org/10.1006/nimg.2001.0978

[14] H. Zou and T. Hastie, “Regularization and variable selection via the elastic net,” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 67, no. 2, pp. 301–320, 2005. [Online]. Available: https://doi.org/10.1111/j.1467-9868.2005.00503.x

[15] Autism Brain Imaging Data Exchange, “ABIDE II release and collection documentation,” https://fcon_1000.projects.nitrc.org/indi/abide/abide_II.html, 2017, accessed 14 August 2026.

[16] 1000 Functional Connectomes Project, “Locally linear embedding preprocessed repository,” https://fcon_1000.projects.nitrc.org/indi/abide/LLE_home.html, 2023, accessed 14 August 2026.